P05

Quorum Sensing and Collective Decision-Making in Honeybees

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Domain DOM_BIO

Abstract

Honeybee swarms choose among prospective nesting sites using waggle dances that represent site quality. When scout bees observe that visitor numbers at a site surpass a quorum threshold, they initiate piping signals that coordinate liftoff without any central census.

Abstract quoted from T. D. Seeley et al., Science (2004).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?